

A BEGINNER'S FIELD GUIDE

Microtuning with M-Suite

How to hear, understand, and compose music beyond twelve equal notes

NO THEORY REQUIRED This guide starts with sound—not notation. You will learn only the theory needed to make deliberate musical choices in mdrone, mkeys, mchord, and mrage.

M-Suite • July 2026

Contents

Part	Topic
1	Start here: what microtuning means
2	The minimum music theory
3	How pitch becomes a tuning system
4	Meet the M-Suite microtuning instruments
5	First piece: a tuned drone in mdrone
6	Second piece: melody in mkeys
7	Harmony in mchord
8	Generative melody with mraga
9	Four creative recipes
10	MIDI, Scala files, saving, and sharing
11	Troubleshooting and ear training
A	Glossary and quick reference

SUGGESTED ROUTE New to music? Read Parts 1–5 in order. Already comfortable with scales and chords? Skim Parts 1–3, then jump to the instrument you want.

How to use this guide

Microtuning is easier to understand with your ears than with equations. Keep an M-Suite instrument open while reading. Whenever the guide gives a number in cents or a ratio, treat it as a label for a sound—not a test you must pass.

- Headphones or decent speakers help you hear slow beating and low drones.
- Keep the master volume low when stacking sustained tones.
- Save or share any scene you like before making a large change.
- The web instruments may evolve; control names in this guide reflect the July 2026 builds.

1. Start here: what microtuning means

The simple definition

Most pianos and software instruments divide an octave into twelve equal steps. This system is called twelve-tone equal temperament, shortened to 12-TET. Microtuning means using a different map of pitch: some notes may sit slightly above or below piano notes, an octave may contain more or fewer than twelve degrees, or the pattern may repeat at an interval other than an octave.

IN EVERYDAY LANGUAGE Microtuning is choosing a different set of pitch colours. It does not automatically mean “tiny intervals,” “wrong notes,” or difficult mathematics.

Why it can sound unfamiliar

Your ear learns expectations through repetition. If you have mainly heard 12-TET, its intervals feel neutral. Another tuning may initially seem bent, sour, luminous, weightless, rough, or strangely calm. Those impressions are useful. Microtuning turns “in tune” from a fixed law into a relationship among notes, timbre, and musical context.

What microtuning changes—and what it does not

Changes	Does not have to change
The exact pitch of scale degrees	Your rhythm or tempo
The colour and beating of chords	Your ability to use repetition
Which intervals feel settled or tense	The need for a memorable musical idea
How instruments agree on pitch	The usefulness of dynamics, texture, and form

A first listening experiment

1. Open mdrone and press HOLD at a comfortable volume.
2. Choose Equal (12-TET). Listen for ten seconds without changing anything.
3. Switch to Just 5-limit, then $\frac{1}{4}$ -comma Meantone, then Slendro.
4. Describe each with ordinary words: smooth, restless, hollow, vocal, metallic, dark. There is no wrong answer.
5. Return to 12-TET. Notice that the “normal” tuning now has its own character.

2. The minimum music theory

Sound, pitch, and frequency

A repeating vibration is measured in hertz (Hz), meaning cycles per second. Faster vibration usually sounds higher. A tone at 440 Hz is the concert A above middle C. A tone at 880 Hz vibrates twice as fast and sounds like the same note one octave higher.

Idea	Plain meaning	Example
Pitch	How high or low a sound feels	A bass note feels lower than a flute note
Frequency	The measured vibration rate	440 Hz
Octave	A 2:1 frequency relationship	440 Hz → 880 Hz
Tonic / root	The pitch that feels like home	A drone centred on C
Scale degree	One position in a chosen pitch set	The fifth step above the tonic

Notes, scales, and intervals

A note name such as C is a label. A scale is an ordered collection of pitches. An interval is the distance between two pitches. Melody presents pitches over time; harmony lets pitches sound together. A drone is one sustained pitch—or a small stable group—that makes every other pitch easy to compare against it.

WHY DRONES MATTER A tonic drone is an audible ruler. When another note sounds above it, you can hear whether the relationship locks, beats, floats, or rubs.

Cents: a ruler for pitch distance

The octave is divided into 1,200 cents. In 12-TET, each semitone is exactly 100 cents. Cents measure distance, not absolute pitch: 700 cents above any starting note is the 12-TET fifth.

Interval in 12-TET	Cents	Beginner image
Semitone	100	One adjacent piano key
Whole tone	200	Two semitones

Interval in 12-TET	Cents	Beginner image
Major third	400	Bright major-chord colour
Perfect fourth	500	Open, suspended colour
Perfect fifth	700	Very stable over a drone
Octave	1200	Same note class, doubled frequency

Ratios: another way to describe intervals

A ratio compares vibration rates. The octave is 2/1; the upper note vibrates twice for every cycle of the lower. A pure fifth is 3/2. A pure major third is 5/4. Small-number ratios often align their overtones and can sound fused or beatless with steady timbres.

12-TET slightly adjusts most ratios so the same keyboard pattern works in every key. Just intonation does the opposite: it prioritises especially clean relationships to a chosen tonic. Neither is universally better; each solves a different musical problem.

Consonance, dissonance, and beating

Consonance usually means a stable or fused relationship; dissonance means tension or complexity. These are perceptions, not moral rankings. Two close frequencies create beating—a rise and fall in loudness caused by interference. Slow beating can animate a drone; fast beating can add roughness. M-Suite's sustained voices make this especially audible.

3. How pitch becomes a tuning system

12-TET: the familiar compromise

Twelve equal semitones make transposition simple: a chord keeps the same shape in every key. The price is that most intervals are slightly displaced from pure ratios. Because the displacement is consistent, most listeners stop noticing it.

Just intonation

Just intonation builds important notes from frequency ratios such as 3/2 and 5/4. It can produce strikingly still chords over a fixed tonic. Change the tonic, however, and the same set of pitches may no longer form pure relationships. In practical composition, treat the drone or root as the centre of gravity.

Meantone and well temperaments

Meantone tunings favour sweet thirds by distributing tuning error unevenly. Historical well temperaments also distribute error unevenly, allowing all keys while giving each one a different colour. In M-Suite, this is why the tuning anchor matters: Follow key re-roots the table on the current key, while Fixed C preserves a fixed historical map.

Equal divisions: EDO and TET

EDO means equal divisions of the octave. 19-EDO divides the octave into nineteen equal steps; 31-EDO uses thirty-one. M-Suite's curated 19-TET and 31-TET tables are twelve-note selections from those larger grids, while mkeys can load arbitrary full scales through Scala files. A name such as "19-TET" therefore does not always mean the visible instrument exposes all nineteen degrees.

Maqam, raga, pelog, and slendro

Many musical traditions organise pitch through more than a fixed cents list: characteristic phrases, intonation, ornaments, hierarchy, rhythm, and performance practice matter. A tuning preset is an entry point, not a complete representation of a culture. Use names respectfully, listen to expert performers, and avoid assuming that selecting "Rast," "Bayati," "Yaman," "Pelog," or "Slendro" automatically makes a piece belong to that tradition.

Non-octave scales

Some systems repeat at something other than 1,200 cents. Bohlen–Pierce, for example, repeats at a 3:1 "tritone." mkeys supports arbitrary scale lengths and non-octave periods through .scl/.kbm import. mdrone's editor is designed around a thirteen-slot table whose final slot is the period; other M-Suite instruments have their own constraints.

System	What to listen for	Good first use
Just 5-limit	Stillness and overtone fusion	A tonic drone with a fifth and third
$\frac{1}{4}$ -comma meantone	Sweet thirds; uneven key colour	Slow chord changes
Maqam Rast / Bayati	Inflected degrees around a tonal centre	Short motifs over a drone
Slendro / Pelog	Uneven spacing and distinctive contour	Sparse percussion plus melody
19/31-EDO-derived tables	Familiar shapes with shifted colour	Reharmonising a simple loop

System	What to listen for	Good first use
Custom / Scala	A personally designed pitch world	Focused experiments

4. Meet the M-Suite microtuning instruments

M-Suite is most powerful as a chain of specialised instruments. You do not need every app in every piece.

Instrument	Role	Microtuning scope
mdrone	Create the tonal centre and long-form texture	Built-in and authored tables; 13-degree editor; per-interval detune; scene links
mkeys	Perform melodies, chords, and arpeggios	Arbitrary scale lengths; non-octave periods; .scl and .kbm import; exact internal tuning
mchord	Design and sequence chord progressions	Twelve-note tunings and 12-tone .scl import; selectable tuning anchor; internal synth only
mrage	Conduct a generative struck melody	Reads tonic and cents from an mdrone link; arbitrary-length imported tables; MPE/pitch-bend MIDI

A USEFUL DIVISION OF LABOUR Let mdrone establish “home,” use mkeys for direct playing, mchord for harmonic sketches, and mrage for autonomous melodic motion.

Important compatibility notes

- mdrone is the canonical source of the shared tuning library used by the family.
- mchord preserves its normal harmony logic and retunes only the final pitch-to-frequency stage. Its microtuning applies to the internal synth; MIDI out remains 12-TET.
- mkeys resolves incoming MIDI notes through the active tuning or .kbm map.
- mrage can output microtonal MIDI as the nearest MIDI note plus pitch bend. MPE mode gives overlapping notes independent bends.
- Live tuning changes in mdrone do not automatically update mrage. Paste a new mdrone scene link after changing the tuning.

5. First piece: a tuned drone in mdrone

Exercise: one centre, three colours

1. Open mdrone, choose a tonic and octave, and press HOLD.
2. Start with one or two clear voices, such as REED and TANPURA. Keep effects restrained so pitch relationships remain audible.
3. In ADVANCED, select Equal (12-TET). Choose a simple relation such as tonic–fifth or drone triad.
4. Listen, then switch to Just 5-limit. Notice whether the beating, body, or apparent brightness changes.
5. Try $\frac{1}{4}$ -comma Meantone and mdrone Signature. Do not search for “best”; choose the colour that supports your mood.
6. Use each displayed degree’s audition control. Read its cents value; in compatible just tunings, notice the ratio label.
7. If the tuning offers a suggested voicing, use APPLY VOICING only after listening to the current relation. Compare deliberately.
8. Add WEATHER, EVOLVE, or a slow effect only after the pitch structure feels clear.

Using fine detune musically

The ± 25 -cent fine-detune control is not merely a correction tool. A deviation of two to five cents can create a slow pulse between sustained tones. Larger offsets become an expressive bend. Keep at least one stable reference pitch, or the whole texture may feel unfocused.

Using ATTUNE

ATTUNE chooses a drone-friendly tuning and relation, then adds gentle detuning. Treat it as a discovery tool: fire it, listen for at least twenty seconds, and keep only results you can describe. ATTUNE is temporarily disabled while the engine warms and has a short cooldown to avoid stacked retunes.

Designing a custom table

1. Begin from a known table rather than twelve random values.
2. Keep degree 1 at 0 cents: this is the tonic.
3. Change one or two degrees. Move a third from 400 toward about 386 cents, or a fifth from 700 toward about 702 cents.
4. Audition each change against the drone. Use words before numbers.

5. Name and save the table. M-Suite prevents silent name collisions and warns before deleting a tuning referenced elsewhere.

BEGINNER RULE Make one pitch decision at a time. If tuning, voicing, timbre, effects, and motion all change together, you cannot learn what caused the result.

6. Second piece: melody in mkeys

Exercise: turn a tuning into a phrase

1. Choose the same tonic and tuning you used in mdrone—or load a matching Scala file.
2. Play only three degrees: tonic, one contrasting degree, and a stable return note. Repeat a short rhythm.
3. Add a fourth degree only when the three-note motif feels memorable.
4. Use the arpeggiator or sequencer to repeat the phrase. Repetition teaches the ear what “home” means in this tuning.
5. Change register before changing tuning. A pitch relationship can feel very different in the bass and treble.
6. Save the session or copy a share link when the phrase and sound work together.

Scala .scl and .kbm files

A .scl file describes the pitch distances in a scale. Values may be written as cents or ratios. A .kbm file describes how computer or MIDI keys map onto those degrees. The scale answers “what pitches exist?”; the keyboard map answers “which key plays which pitch?”

WHY A KEYBOARD MAP MATTERS A 19-note or non-octave scale does not naturally fit a twelve-key repeating keyboard. A .kbm map makes the relationship explicit instead of forcing a misleading piano layout.

Composing with an unfamiliar scale

- Find the tonic and one stable neighbour by ear.
- Limit the first motif to three or four degrees.
- Repeat the same contour at a new register.

- Use rhythm, silence, and timbre as familiar anchors.
- Avoid filling every available degree merely because it exists.

7. Harmony in mchord

What stays familiar

mchord builds chord progressions with familiar harmonic categories and voice-leading modes. Microtuning does not rewrite that engine; it adjusts the final frequency of each of the twelve pitch classes. This makes it a friendly bridge from conventional harmony into new interval colour.

The tuning anchor

Anchor	Meaning	Use it when...
Follow key	The tuning table moves so the current key is its tonic	Working with just intonation, maqam-like scales, or frequent key changes
Fixed C	The table remains rooted on C	Exploring historical well temperaments and key colour
Fixed note	The table remains rooted on a chosen pitch	Building a piece around a pedal point or external fixed reference

Exercise: one progression, three temperaments

1. Create a slow four-chord progression in 12-TET. Choose Smooth voice leading.
2. Switch to Just 5-limit with Follow key. Listen to held thirds and fifths.
3. Switch to Werckmeister III with Fixed C. Transpose the progression through several keys and notice the changing colour.
4. Try a world or EDO-derived table. Reduce chord density if the result becomes crowded.
5. Record or save the version whose tension supports the form—not simply the most exotic tuning.

A practical limitation

mchord accepts twelve-note tunings and twelve-tone Scala imports. Its internal synth is microtuned, but its MIDI output remains 12-TET. To capture the exact tuning, record mchord's audio. If an external synth must receive microtonal notes, use a separate compatible workflow such as mraga's pitch-bend/MPE output or a synth that loads the same Scala scale directly.

8. Generative melody with mraga

The core workflow

1. Build and save a drone in mdrone.
2. Press LINK to copy its self-contained scene URL.
3. Open mraga and paste the mdrone link. mraga reads the exact tonic, scale cents, and repeat period.
4. Start the improviser. Begin with low DENSITY, moderate SILENCE, calm RESTLESS, and fairly high FOCUS.
5. Choose REGISTER and VOICE to sit above the drone without masking it.
6. Raise THEME when you want recurring motifs; lower FOCUS when you want a wider pitch palette.
7. Reroll the seed only after deciding whether the current behaviour—not merely one lucky phrase—supports the piece.

What the seven conducting controls mean

Control	In ordinary language
DENSITY	How often notes happen
REGISTER	Where the melody sits from low to high
RESTLESS	How far it wanders and how strongly it returns home
SILENCE	How much breathing space appears
RHYTHM	How loose or grid-like the timing feels
THEME	How strongly a motif repeats
FOCUS	How narrow or wide the note palette is

Why mraga sounds connected to the drone

Its phrases are drawn from the imported tuning and lean toward resting degrees near the tonic, fifth, fourth, and strong thirds. It uses directed contours, motif repetition, breathing, and glides rather than a note-by-note random walk. Named raga features add grammar beyond the tuning, but they still deserve contextual listening and study.

Microtonal MIDI

Standard MIDI notes name piano-key pitches; pitch bend supplies the offset. In Single Channel mode, one bend affects all notes on that channel, so overlapping notes can retune each other. MPE rotates notes across channels 2–8, allowing independent bends. Set the receiving synth to a ± 2 -semitone pitch-bend range and enable MPE if available.

9. Four creative recipes

Recipe A — Still just-intonation ambient

- mdrone: Just 5-limit; tonic–fifth or drone triad; REED + AIR; slow WEATHER movement.
- mkeys: three-degree motif with long notes and wide rests.
- Form: introduce one degree every 30–60 seconds; remove material before adding effects.

Recipe B — Meantone chord study

- mchord: $\frac{1}{4}$ -comma Meantone; slow progression; Smooth or Close voicing.
- Compare Follow key with Fixed C. Let one chord ring long enough to hear its beating.
- Record internal audio to preserve the tuning.

Recipe C — Drone and conducted raga line

- mdrone: choose a suitable tuning and a sparse tanpura/reed texture; copy the scene link.
- mraga: paste the link; low DENSITY, high SILENCE, calm RESTLESS; select a raga grammar only after reading its musical context.
- Use free timing for an opening; introduce BPM/Link later for a stronger pulse.

Recipe D — Non-octave miniature

- mkeys: import a non-octave .scl and its .kbn map.
- Use only four degrees. Write a repeated rhythmic cell before exploring the full scale.
- Avoid conventional chord labels; think in contour, register, density, and spectral colour.

10. MIDI, Scala files, saving, and sharing

Choose the right transport

Goal	Recommended method
Preserve the exact sound	Record/export audio from the instrument
Let another M-Suite user reopen the scene	Use the app's self-contained share link
Archive/edit the scene data	Export the app's JSON/session format where available
Move a tuning between compatible tools	Use .scl; add .kbm when keyboard mapping matters
Drive an MPE synth from mraga	MPE MIDI with matching ± 2 -semitone bend range
Send mchord to ordinary MIDI gear	Expect 12-TET MIDI; record internal audio for its microtuning

A safe sharing checklist

- Include the tonic, reference frequency if nonstandard, tuning name, and whether the tuning follows the key or stays fixed.
- For custom scales, share the .scl and .kbm files as well as audio.
- For mraga, share both the source mdrone scene and the mraga scene/seed when reproducibility matters.
- Render a short audio example so collaborators know the intended result before configuring their tools.

11. Troubleshooting and ear training

Symptom	Likely cause	Try this
Everything sounds “wrong”	Too many unfamiliar variables changed together	Return to a tonic drone; use three degrees; remove effects
Chords sound muddy	Dense voicing, low register, or fast beating	Raise voicing, remove a note, simplify timbre

Symptom	Likely cause	Try this
mrage no longer matches mdrone	mdrone tuning changed after the link was pasted	Copy and paste a fresh mdrone link
External synth ignores tuning	It received ordinary MIDI or wrong bend range	Use MPE/pitch bend; set ± 2 semitones; verify channel mode
Overlapping MIDI notes bend together	Single-channel pitch bend limitation	Use MPE or avoid overlapping releases
Scala scale feels scrambled	Keyboard map missing or unsuitable	Load/create a matching .kbm map
The music feels like a demo	No form, contrast, or memorable motif	Limit materials; repeat; introduce one clear change per section

A seven-day ear-training plan

1. Day 1: compare 12-TET and Just 5-limit over one tonic.
2. Day 2: identify slow versus fast beating between sustained notes.
3. Day 3: write a three-note motif in Slendro or Pelog.
4. Day 4: compare Follow key and Fixed C in mchord.
5. Day 5: move one custom degree by 5, 10, and 25 cents.
6. Day 6: import one Scala scale into mkeys and map it deliberately.
7. Day 7: make a two-minute piece with a beginning, change, and ending.

Appendix A — Glossary and quick reference

Term	Meaning
12-TET	The octave divided into twelve equal 100-cent steps
Cent	1/1,200 of an octave; a unit of pitch distance
EDO	Equal divisions of the octave
Frequency	Vibration rate measured in hertz
Interval	Distance between two pitches

Term	Meaning
Just intonation	Tuning based on selected whole-number frequency ratios
Microtuning	Any deliberate pitch organisation beyond the assumed standard tuning
MPE	MIDI mode giving notes separate expressive channels, including pitch bend
Period	The interval after which a scale pattern repeats
Ratio	A comparison of two frequencies, such as 3/2
Scala .scl	Text format describing a scale's pitch distances
Scala .kbm	Text format mapping keys/MIDI notes to scale degrees
Scale degree	A pitch position in a scale
Tonic	The pitch perceived as home or reference

Quick-reference formulas

You do not need these to compose, but they explain what the software calculates:

- Frequency from cents: $\text{new frequency} = \text{tonic} \times 2^{(\text{cents} \div 1200)}$.
- Cents from a ratio: $\text{cents} = 1200 \times \log_2(\text{ratio})$.
- One octave above: $\text{frequency} \times 2$. One octave below: $\text{frequency} \div 2$.

M-Suite links

- [mdrone](#) — microtonal drone and tuning authoring
- [mkeys](#) — performance, arbitrary scales, Scala import
- [mchord](#) — twelve-note harmonic exploration
- [mraga](#) — generative melody linked to mdrone

Final thought

Microtuning is not a contest to use the strangest scale. It is the craft of making pitch relationships intentional. Start with a centre, listen long enough to hear the relationships, limit your materials, and let repetition teach both you and the listener what this particular musical world means.